

Fall 2025: CS 141 Exam #2

This is an **open note** exam. You are permitted to use written or printed notes, but not a computer. The last page is a reference for the APIs that we've used in class; we suggest that you tear it off to make it easier to use during the exam. Students whose native language is not English may use a dictionary. *Nothing else is permitted.*

If a problem is unclear to you, include a brief explanation of how you interpret it.

Show your work!

You will not get partial credit if we cannot figure out how you got your answer.

Do not talk to other students, or look at other students' exams.

Please sign below to indicate that you have upheld the highest standards of Knox College's Honor Code, and that you have not cheated in any way.

1. **Types (20 points).** Suppose we have the following variable declarations.

```
double[] prices;  
int[][] table;  
char[] letters;  
String[][] names;
```

What is the **type** of the following expressions?

a. prices

b. prices[2]

c. table

d. table[0]

e. table[0][1]

f. table[0][1] + 1.5

g. letters

h. letters[1]

i. "Hello" + letters[1]

j. names

k. names[0]

l. names[table[0][1]]

m. names[0][2]

n. names[0][2].charAt(0)

2. **Method signatures (20 points).** Write signatures for each of the following potential methods. We do not need any of the code; just the signatures. Assume we are creating a Hero class for a Legend of **Zelda** style game.
- a. An `isAlive` method in the `Hero` class that tells us whether the hero is still alive.
 - b. A `getCurrentRegion` method in the `Hero` class that returns the name of the region (as a `String`) where the hero is currently exploring.
 - c. A `getCompanion` method in the `Hero` class that returns the `Companion` object currently traveling with the hero.
 - d. A `useItem` method in the `Hero` class that takes an `Item` object as a parameter and returns a `boolean` indicating whether the item was successfully used.
 - e. A constructor for the `Hero` class that takes a `String` for the hero's name, an `int` for starting hearts, and a `Weapon` for the hero's initial weapon.

3. **Testing to find errors (20 points).** Each of the following functions is syntactically correct, but contains at least one logic error. The error could be that the function produces the wrong result (i.e. it says it return the max but it sometimes returns the wrong value), or it could be that it crashes for some inputs (i.e. sometimes you get `ArrayIndexOutOfBoundsException` or `NullPointerException`). Explain what the error is, and how to fix it.

a.

```
// return the sum of all positive values in the array
public static int sumPositives(int[] nums) {
    int sum = 0;
    for (int i = 1; i <= nums.length; i++) {
        if (nums[i] > 0) {
            sum += nums[i];
        }
    }
    return sum;
}
```

b.

```
// return a new array containing only the even numbers in nums
public static int[] filterEvens(int[] nums) {
    int count = 0;
    for (int i = 0; i < nums.length; i++) {
        if (nums[i] % 2 == 0) {
            count++;
        }
    }
    int[] result = new int[count];
    for (int i = 0; i < nums.length; i++) {
        if (nums[i] % 2 == 0) {
            result[i] = nums[i];
        }
    }
    return result;
}
```

c.

```
// return the array rotated one spot to the right, wrapping around
// So [1, 2, 3, 4] becomes [4, 1, 2, 3]
public static int[] rotateRight(int[] nums) {
    for (int i = nums.length - 1; i > 0; i--) {
        nums[i] = nums[i-1];
    }
    nums[0] = nums[nums.length-1];
    return nums;
}
```

d.

```
public static boolean sameNegatives(int[] arr1, int[] arr2,
int[] arr3) {
    int count1 = 0;
    int count2 = 0;
    int count3 = 0;
    for (int i = 0; i < arr1.length; i++) {
        if (arr1[i] < 0)
            count1++;
    }
    for (int i = 0; i < arr2.length; i++) {
        if (arr2[i] < 0)
            count2++;
    }
    for (int i = 0; i < arr3.length; i++) {
        if (arr3[i] < 0)
            count3++;
    }

    return (count1 + count2 + count3) % 3 == 0;
}
```

4. **Arrays (20 points).** Write a function (static method) called `countStartsWith` that takes an array of `String` and a single `char` letter as parameters, and returns the number of words that start with that letter. For example:

Sample Input	Expected Output
<code>countStartsWith(["apple", "banana", "apricot", "avocado"], 'a')</code>	3
<code>countStartsWith(["pikachu", "bulbasaur", "big bulba", "meowth"], 'b')</code>	2
<code>countStartsWith([], 'x')</code>	0

```
public static int countStartsWith(String[] words, char letter) {
```

```
}
```

5. **Grids (20 points).** What values are stored in the array after this code runs? Write the contents of the 2D array.

```
int[][] arr = new int[5][5];

for (int r = 0; r < arr.length; r++)
{
    for (int c = 0; c < arr[0].length; c++)
    {
        if (r == c)
            arr[r][c] = 1;
        else if (r + c == arr.length - 1)
            arr[r][c] = 2;
        else
            arr[r][c] = 0;
    }
}
```
